



Terraform Engineering Bootcamp

Week 1 Student Lab

Terraform Foundations

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Lab Objective

By the end of this lab, you will be able to:

- ✓ Create a Terraform project
 - ✓ Configure a Terraform provider
 - ✓ Initialize Terraform
 - ✓ Create your first resource
 - ✓ Format and validate Terraform code
 - ✓ Preview infrastructure changes
 - ✓ Deploy infrastructure
 - ✓ Understand Terraform State
 - ✓ Modify infrastructure
 - ✓ Safely destroy infrastructure
-

Prerequisites

Ensure the following software is installed.

Verify Terraform

```
terraform version
```

Expected Output

```
Terraform v1.x.x
```

Verify Git

```
git --version
```

Expected Output

```
git version 2.x.x
```

Open VS Code

```
code .
```

LAB 1 — Create Your Terraform Project

Create a new folder.

```
mkdir terraform-foundations
```

Move into the folder.

```
cd terraform-foundations
```

Open Visual Studio Code.

```
code .
```

Expected Folder

```
terraform-foundations
```

```
(empty)
```

□ LAB 2 — Create Your First Terraform File

Create a new file called

`main.tf`

Copy the following code into **main.tf**

```
terraform {  
  required_providers {  
    local = {  
      source  = "hashicorp/local"  
      version = "~> 2.5"  
    }  
  }  
}  
  
provider "local" {}
```

Save the file.

Knowledge Check

What is Terraform?

Terraform is an Infrastructure as Code (IaC) tool used to create and manage infrastructure.

What is a Provider?

A provider allows Terraform to communicate with a specific platform.

Which provider are we using?

Local Provider

□ LAB 3 — Initialize Terraform

Run:

```
terraform init
```

Expected Output

```
Terraform has been successfully initialized!
```

You should now see:

```
terraform-foundations
|
|— main.tf
|— .terraform/
|— .terraform.lock.hcl
```

□ LAB 4 — Create Your First Resource

Add the following code below the provider block.

```
resource "local_file" "welcome" {
    filename = "welcome.txt"

    content = <<EOF
Welcome to Terraform Engineering

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Instructor:
Mthulisi Zulu

EOF
}
```

Save the file.

□ LAB 5 — Format and Validate Code

Format your code.

```
terraform fmt
```

Validate your code.

```
terraform validate
```

Expected Output

```
Success!  
The configuration is valid.
```

□ LAB 6 — Preview Infrastructure

Run:

```
terraform plan
```

Look for

```
+ create
```

Question

Has Terraform created the file?

Answer

No.

Terraform is only showing what **will** happen.

□ LAB 7 — Deploy Infrastructure

Run

```
terraform apply
```

When prompted, type

yes

Expected Output

Apply complete!

Check your folder.

You should now see

welcome.txt

Open the file and verify its contents.

❑ LAB 8 — Understanding Terraform State

Look inside your project folder.

You should now see

terraform.tfstate

What is Terraform State?

Terraform State stores information about everything Terraform creates.

Terraform uses this file to keep track of your infrastructure.

⚠ **Never edit this file manually.**

❑ LAB 9 — Add Another Resource

Add the following resource to **main.tf**

```
resource "local_file" "cloud_goals" {  
    filename = "cloud-goals.txt"  
  
    content = <<EOF  
My Cloud Goals
```

1. Learn Terraform
2. Learn AWS
3. Build Automation Skills
4. Become a Cloud Engineer

EOF

}

Run

```
terraform fmt
terraform validate
terraform plan
terraform apply
```

Expected Files

welcome.txt

cloud-goals.txt

☐ LAB 10 — Update Infrastructure

Modify the content of **cloud-goals.txt** inside **main.tf**

Add

5. Learn Cloud Governance
 6. Master Infrastructure as Code
-

Run

```
terraform plan
```

Notice

```
~ update
```

This means Terraform detected a change.

Apply the update.

```
terraform apply
```

LAB 11 — Destroy Infrastructure

Preview what will be destroyed.

```
terraform plan -destroy
```

Now destroy everything.

```
terraform destroy
```

Type

```
yes
```

Expected Output

```
Destroy complete!
```

The files created by Terraform should now be removed.

Knowledge Check

Answer the following questions.

1. What command initializes Terraform?

2. What command checks your Terraform syntax?

3. What command shows what Terraform will do?

4. What command creates infrastructure?

5. What command removes infrastructure?

6. What file stores Terraform state?

Week 1 Challenge

Create **one additional Terraform resource** that creates a file called

`career-roadmap.txt`

The file should contain your personal cloud career goals.

Requirements

- Use the **Local Provider**
 - Create the file using **Terraform only**
 - Do **not** create the file manually
-

Submission Checklist

Submit the following:

- ☐ `main.tf`
 - ☐ Screenshot of `terraform init`
 - ☐ Screenshot of `terraform plan`
 - ☐ Screenshot of `terraform apply`
 - ☐ Screenshot showing the created files
 - ☐ Screenshot of `terraform destroy`
-

Congratulations!

You have completed **Week 1 – Terraform Foundations**.

You now understand the basic Terraform workflow:

Write Code
↓
terraform init
↓
terraform fmt
↓
terraform validate
↓
terraform plan
↓
terraform apply
↓
Infrastructure Created
↓
terraform destroy

Next Week: Working with Multiple Resources, Variables, Outputs, and Git Version Control.